

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High Section
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

REGISTRATION
NUMBER

BIOLOGY

Paper 2 Core Paper

9747/02

September 2009

2 hours

Additional Materials: **Answer Paper**

READ THESE INSTRUCTIONS FIRST

Write your Biology class, registration number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.

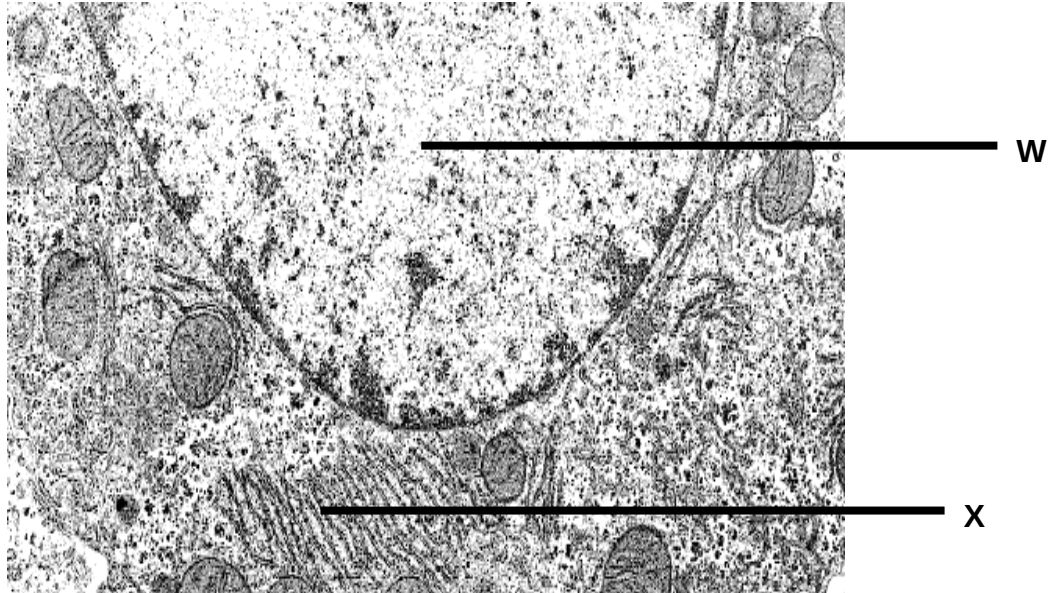
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/ 10
2	/ 12
3	/ 12
4	/ 12
5	/ 10
6	/ 12
7	/ 12
Section B	
8 / 9	/ 20
TOTAL	/ 100

Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 is an electron micrograph of a eukaryotic cell.



[Picture adopted from: <http://www.ucmp.berkeley.edu/alllife/eukaryotamm.html>]

Fig. 1.1

- (a) Complete the table below by stating **three** ways in which Organelle **W** differs from that of Organelle **X**. [3]

	Organelle W	Organelle X
1	<i>Double membrane bound organelle</i>	<i>Single membrane bound organelle</i>
2	<i>Absence of ribosomes studded on the membrane surface</i>	<i>Presence of ribosomes studded on the membrane surface</i>
3	<i>Presence of genetic material</i>	<i>Absence of genetic material</i>

- (b) Explain how proteins are held in the membrane of Organelle **W**. [2]

- *Hydrophobic / non-polar regions of the proteins consist of one or more stretches of non-polar amino acids*
- *Able to interact / form hydrophobic interaction with the hydrocarbon / fatty acid chains of phospholipid molecules*
- *Hydrophilic regions of the proteins (containing charged / polar R group) are exposed to the aqueous solutions on either side of the membrane*
- *Able to form hydrophilic interactions between the hydrophilic (polar) regions of the amino acids and the hydrophilic head of the phospholipid molecule*

Ehlers-Danlos syndrome is a group of inherited connective tissue disorders, which are caused by a defect in the synthesis of collagen. People with Ehlers-Danlos syndrome have soft, velvety skin that is highly elastic (stretchy) and fragile.

- (c) (i) The tripeptide sequence of glycine-glycine-alanine recurs in collagen.

The structural formula of glycine and alanine are given in Fig. 1.2.

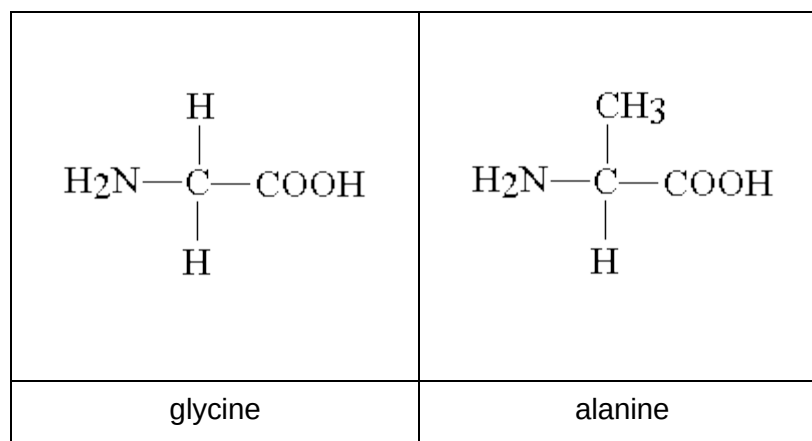


Fig. 1.2

In the space below, draw the structural formula of the tripeptide sequence. Indicate and name the bond formed between these amino acids. [2]

- *Naming of bond (peptide bond)*
- *Correct indication of bond formed*
- *Correct order*
- *Between the amino group of one amino acid and carboxyl group of another*
- *Elimination of water*

- (ii) Explain how the molecular structure of collagen contributes to its physical properties. [2]

<i>Molecular structure</i>	<i>Physical properties</i>
• <i>Each collagen molecule / tropocollagen is made up of 3 polypeptide chains</i> • <i>The 3 polypeptide chains are then wound into a right-handed coiled coil, a triple helix that is stabilized by (numerous) hydrogen bonds</i> • <i>Many of the triple helices are arranged with their ends regularly staggered</i> • <i>So that they can be cross-linked by covalent (aldehyde) bonds</i>	• <i>Organized into long parallel bundles of collagen fibers with great tensile strength</i>

- (iii) Suggest how changing one base in the nucleotide sequence of a human gene that encodes the major component of collagen would result in Ehlers-Danlos syndrome. [1]

- *Missense mutation / base pair insertion / base pair deletion / non-sense mutation*
- *Change in codon*
- *Alter the structure and function of collagen*

[Total: 10]

2 (a) Explain how enzymes lower the activation energy required for a reaction. [3]

- *Hexokinase / enzyme reduces the activation energy required for glucose 6-phosphate formation*
- *When the substrate molecules / glucose and ATP collide with the active site of hexokinase in the right orientation*
- *A enzyme-substrate complex will be formed*
- *Contact residues of enzyme; hold the substrate molecules / glucose and ATP in the right orientation that favours the reaction*
- *Arrangement of substrate molecules in the active site may cause physical stress to the bonds in the substrate molecules; causing them to break more easily*
- *Catalytic residues of enzyme can change the charge on substrate molecules / alter distribution of electrons within the substrate molecules; to increase the reactivity of the substrates*
- *Thus the initial amount of energy required for product / glucose 6-phosphate formation is lower*

Fig. 2.1 shows only part of the multi-step process of glycolysis.

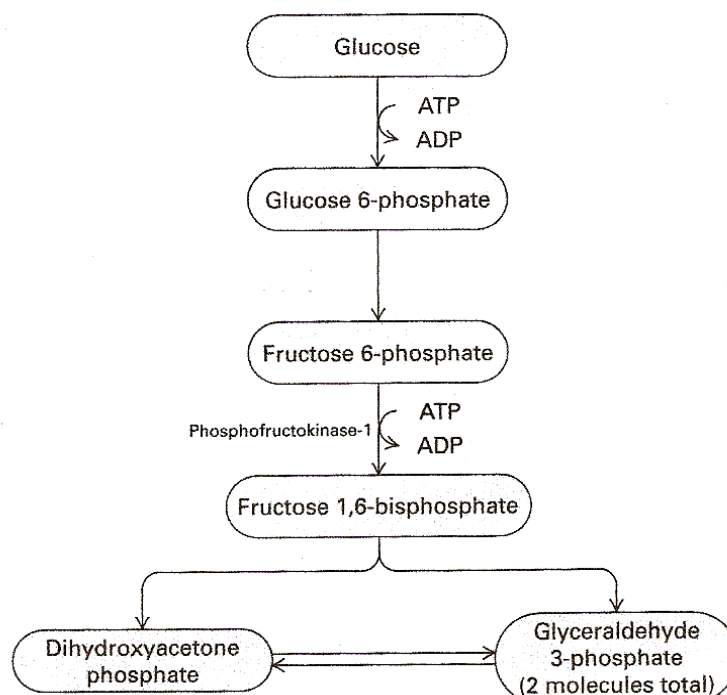


Fig. 2.1

The conversion of fructose-6-phosphate to fructose 1,6-bisphosphate is catalysed by the enzyme, phosphofructokinase-1. A natural inhibitor of phosphofructokinase-1 is citrate, which binds to an allosteric site on phosphofructokinase-1.

- (b) Based on the information given, suggest the significance of the inhibition of phosphofructokinase-1 by citrate in a respiring cell. [2]

- *Feedback inhibition*
- *Regulation of citrate level*
- *Excess citrate causes inhibition of glycolysis*
- *Less pyruvate / less acetyl coA produced*
- *Brings amount of citrate produced by Krebs cycle back to normal*
- *High level of citrate indicates Krebs cycle is at maximum rate and thus glycolysis / earlier reactions of respiration can slow down*

A different group of inhibitors has been found to act by binding to components of the electron transport chains in chloroplasts and mitochondria. These inhibitors affect both photophosphorylation and oxidative phosphorylation.

- (c) Describe **three** differences between photophosphorylation and oxidative phosphorylation. [3]

	<i>Photophosphorylation</i>	<i>Oxidative Phosphorylation</i>
<i>Location</i>	<i>Thylakoid membrane</i>	<i>Inner mitochondrial membrane</i>
<i>Direction of H⁺ pumping / H⁺ movement / H⁺ gradient</i>	<i>H⁺ is pumped from stroma to thylakoid space / lumen, moves back into stroma through ATP synthetase</i>	<i>H⁺ is pumped from matrix to intermembranal space, moves back into matrix through ATP synthetase</i>
<i>Electron donors</i>	<i>Water and photosystems</i>	<i>Respiratory substrates / NADH and FADH₂</i>

Rubisco (RuBP carboxylase) activity can be reduced by its inhibitor, oxygen.

Fig. 2.2 shows the rate of photosynthesis of a plant without inhibition (Graph A) and with inhibition by oxygen (Graph B).

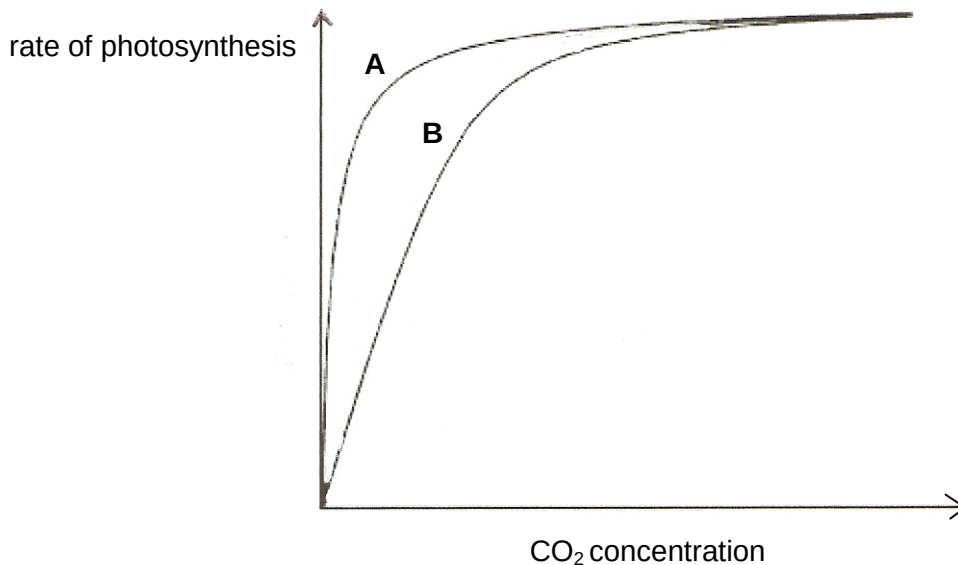


Fig. 2.2

(d) With reference to Fig. 2.2, state, with reasons, whether oxygen is a competitive or non-competitive inhibitor. [2]

- *Competitive inhibitor*
- *Rate of photosynthesis reaches V_{max}*
- *Increasing substrate concentration (CO₂) reduces the effect of the inhibitor as at higher concentrations CO₂ will out-compete O₂ to bind to the active sites*

(e) Account for the decrease in a plant's productivity when Rubisco has been inhibited. [2]

- *Fixation of CO₂ decreases / rate of combination of CO₂ and RuBP decreases*
- *Fewer unstable 6C compounds / GP molecules formed per unit time / ref to rate*
- *Rate of G3P production decreases*
- *Amount of G3P channeled out to form glucose decreases*
- *Thus less glucose formed from photosynthesis*

[Total: 12]

- 3 Fig. 3.1 shows the activation of transcription initiation in eukaryotes by recruitment of the eukaryotic RNA polymerase II holoenzyme complex.

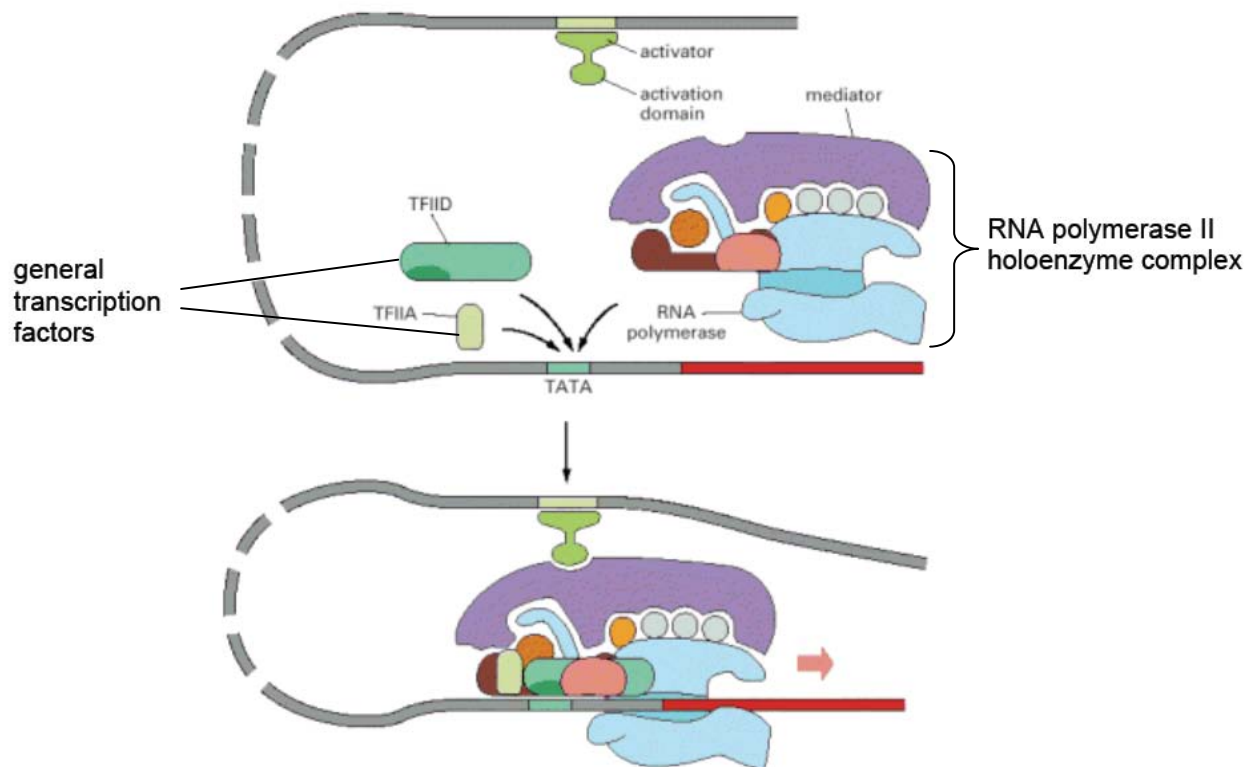


Fig. 3.1

- (a) With reference to Fig. 3.1, state **two** ways in which transcription in eukaryotes differs from that in prokaryotes.

1. *in eukaryotes, transcription begins when transcription factors and RNA polymerase II holoenzyme complex bind to the promoter while in prokaryotes, transcription begins when RNA polymerase binds to the promoter*
2. *in eukaryotes, RNA polymerase (and GTFs) binds to the TATA box while in prokaryotes, RNA polymerase binds to the Pribnow box* [2]

- (b) The product of transcription is RNA.

Account for the roles of **three** types of RNA in producing a particular type of protein.

sequence of bases in mRNA / ref to codons , determine sequence of amino acids

different tRNA for each amino acid / tRNAs bring the correct amino acids to the ribosome for translation

rRNA is a structural component of ribosomes, which are the sites of protein synthesis [3]

Tuberculosis, or TB, is an infectious bacterial disease caused by *Mycobacterium tuberculosis*, which most commonly affects the lungs. It is transmitted from person to person via droplets from the throat and lungs of people with the active respiratory disease.

- (c) TB can be treated with the antibiotic rifampicin, which binds selectively to RNA polymerase in bacteria.

Explain why rifampicin kills the bacteria that cause TB.

Inhibits (initiation of) transcription

no bacterial mRNA synthesised

no bacterial proteins synthesised

bacteria cannot grow / reproduce

[2]

- (d) Explain why mutations for antibiotic resistance spread so rapidly among bacteria.

largely because bacteria reproduce very quickly

mutation can be quickly passed on to large numbers of descendents (usually) via plasmids during binary fission

through conjugation, whereby transfer of antibiotic resistance genes occurs across a mating bridge

through transformation, whereby antibiotic resistance genes taken up by competent bacterial cells through transient pores in the bacterial cell wall

through transduction, whereby (defective) bacteriophages transfer bacterial genes / antibiotic resistance genes

[4]

- (e) TB disease is best treated with combinations of several antibiotics to reduce the risk of the bacteria developing antibiotic resistance.

Suggest **one** reason for this treatment plan.

(antibiotics) target bacteria at different levels

e.g. preventing translation / production of bacterial proteins or synthesis of new cell walls

[1]

[Total: 12]

- 4 Fig. 4.1 shows the sequence of how glucagon enhances glucose output from the liver during fasting by stimulating the transcription of targeted genes via the cyclic AMP-inducible factor, CREB.

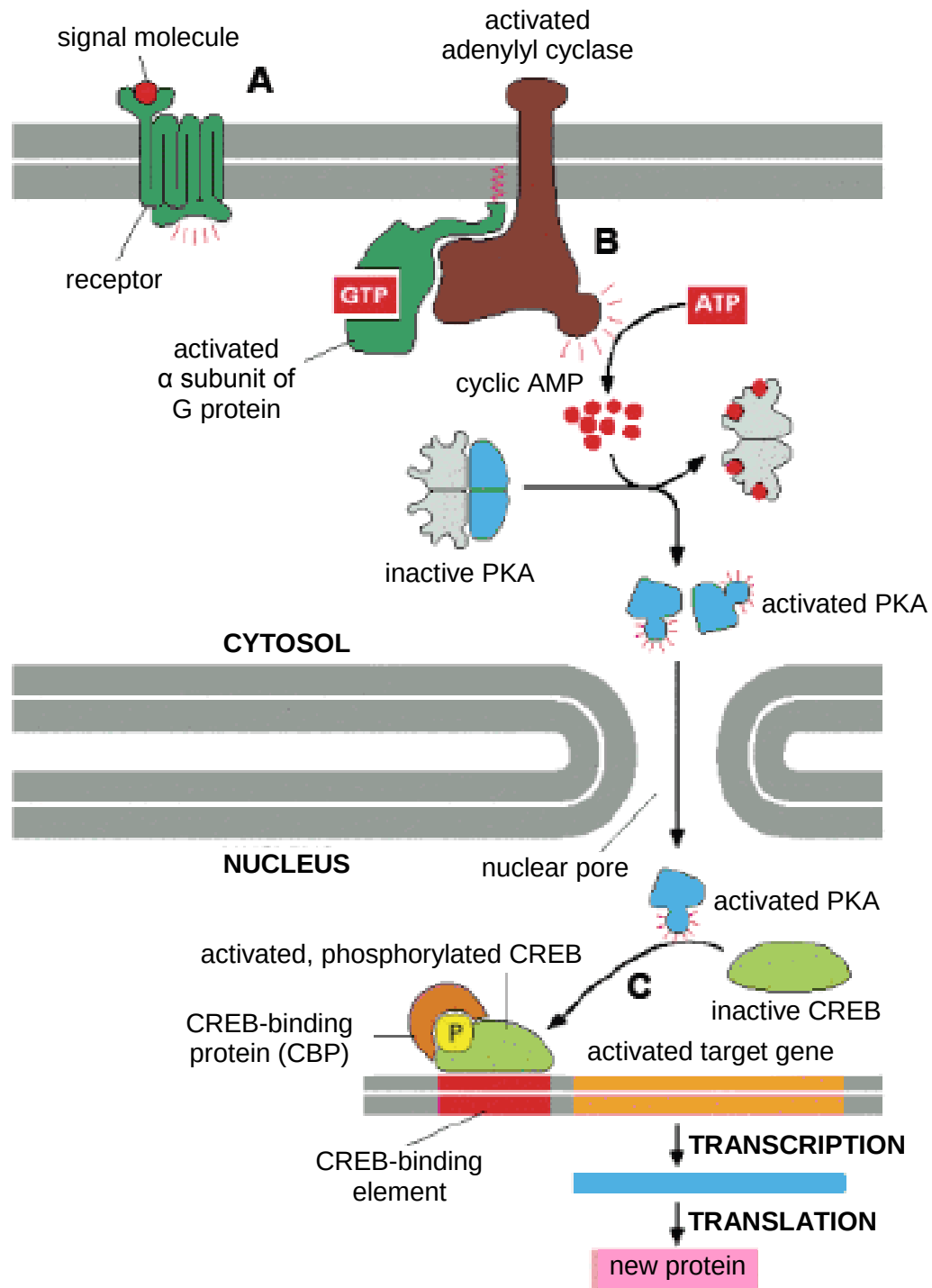


Fig. 4.1

- (a) State the exact site of glucagon secretion in the body.

α cells

islets of Langerhans in the pancreas / pancreatic islets

[1]

(b) Describe the stages **A**, **B** and **C** shown in Fig. 4.1.

A: ligand-receptor interaction

binding of glucagon to GPCR

[1]

B: activation of adenylyl cyclase

by activated G protein

conversion of ATP to cAMP

ref to cAMP as second messengers

activation of inactive PKA by cAMP

[2]

C: activated PKA enters the nucleus

(activated PKA) phosphorylates inactive CREB

recruits CBP to form active CREB

(active CREB) acts as a specific transcription factor / activator

binds to CREB-binding element to activate transcription of targeted gene

[2]

(c) Suggest how glucagon may bring about an increase in the blood glucose concentration.

glucagon bind to receptors on their target (liver) cells

increase glycogenolysis / breakdown of glycogen to glucose by the liver

increase gluconeogenesis / formation of glucose from amino acids and fats

glucose released into the bloodstream restoring normal blood glucose level

[2]

To date, glucagon-binding sites have been identified in multiple tissues, other than the liver.

- (d) Using this information and with reference to Fig. 4.1, explain the significance of the glucagon signaling pathway.

1. allows target cells to respond very quickly to ligand binding at cell-surface receptors / extracellular ligand-binding event

leads to changes in gene transcription in the nucleus / activate genes in the nucleus

2. many different types of cells in the body / multiple tissues can respond to glucagon

because they (all) possess glucagon-binding sites

3. signal amplification

many cAMP / 2nd messengers produced to activate inactive PKA [3]

- (e) Suggest **one** treatment in cases of severe hypoglycemia when the victim is unconscious.

(intravenous) drip / injection of glucose [1]

[Total: 12]

- 5 The coordinated activity of an organism relies upon a continuous input of information from the internal and external environments. Information is in the form of stimuli, which are detected by receptors. One such receptor is called a Pacinian corpuscle, which is found in the skin.

Fig. 5.1 shows the electrical activity recorded by two microelectrodes inserted into:

- the axon within the Pacinian corpuscle, and
- the axon of the sensory neurone leaving the corpuscle.

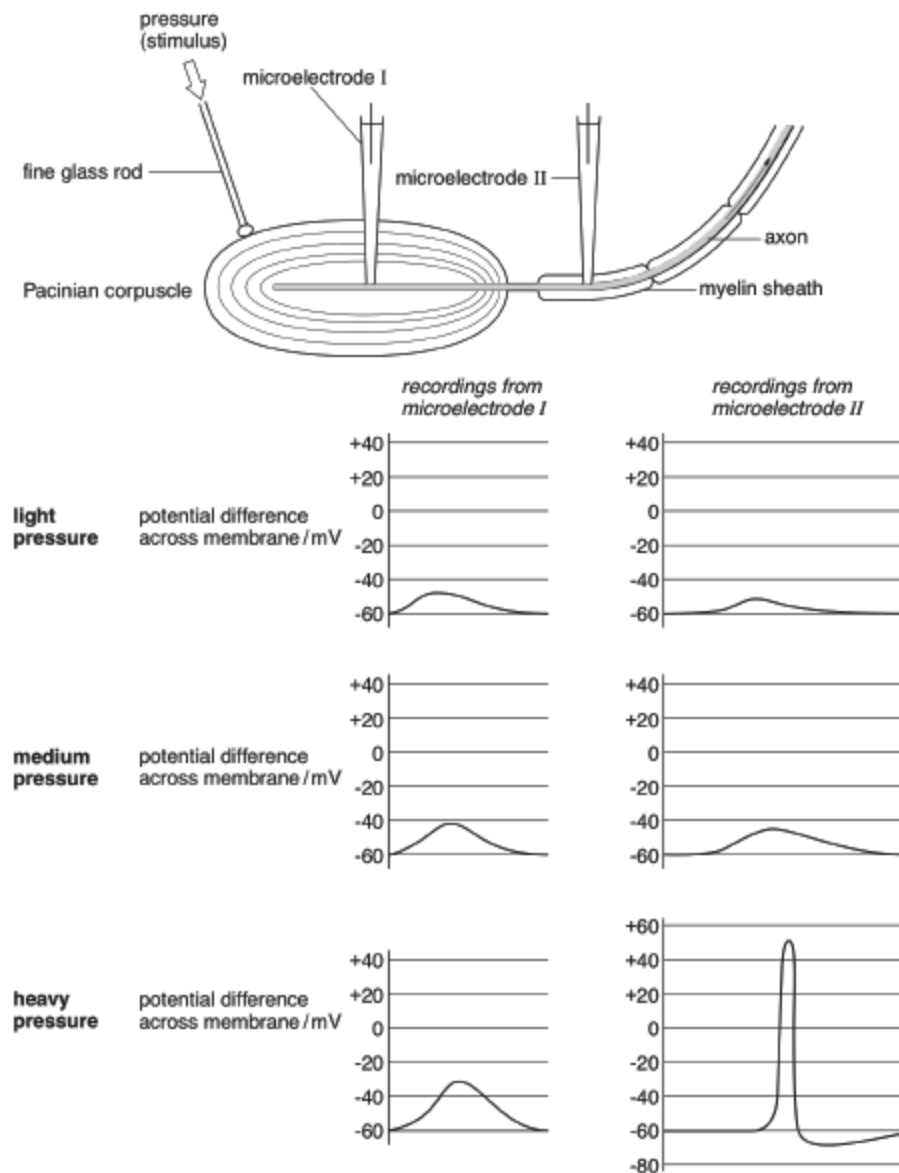


Fig. 5.1

- (a) Using the information in Fig. 5.1, describe and explain the changes in electrical activity in both the receptor and in the sensory neurone as increasing pressure is applied to the Pacinian corpuscle.

applying increasing pressure results in a less negative potential difference (PD) being detected at electrode I

data quoted (within range -50mV to -30 mV from light to heavy pressure)

applying light or medium pressure results in a smaller change in PD

data quoted (within range -55mV to -45mV)

no action potential detected in axon / sensory neurone by electrode II

when heavy pressure is applied, a larger change in PD is detected

PD in electrode I exceeds threshold value (-40mV to -30 mV)

action potential is detected in axon / sensory neurone by electrode II

data quoted for action potential (+45mV to +55 mV)

(less negative PD due to) voltage-gated sodium channels opening

Na⁺ move / diffuse into corpuscle / receptor / axon

causes depolarisation / from -ve to +ve PD

applying increasing pressure causes more voltage-gated sodium channels to open

if sufficient Na⁺ enters to produce adequate depolarization to overcome / exceed the threshold potential

an action potential will be propagated / generated in the sensory neurone / ref to all or none principle [5]

- (b) Transmission of an impulse across a synapse is triggered by a sudden influx of calcium ions into the synaptic knob.

Describe how this influx of calcium ions occurs through the presynaptic membrane.

Depolarization of the presynaptic membrane opens voltage-gated Ca²⁺ channels

The influx of Ca²⁺ ions is due to their diffusion down their concentration / diffusion gradient [2]

- (c) A new painkiller that is much more effective than morphine became available in the United Kingdom in 2006. It was extracted from the venom of the Magician's Cone Snail, *Conus magus*. *C. magus* is a native of coral reefs in the Pacific Ocean. It injects the venom into fish, which are paralysed and swallowed whole. Scientists have discovered that the venom blocks calcium channels at synapses between neurones.

Explain how the blocking of calcium channels can result in both reduction of pain in humans and paralysis in fish.

no uptake of calcium ions into presynaptic membrane / terminal

no movement / fusion of synaptic vesicles to / with presynaptic membrane

no exocytosis / release of neurotransmitter / acetylcholine

no diffusion / movement (of neurotransmitter) across synaptic cleft

no binding (of neurotransmitter) to receptor on postsynaptic membrane

no depolarisation of postsynaptic membrane

[3]

[Total: 10]

- 6 Usually, tomato plants have purple stems, but in some varieties, the purple pigment is missing and the stem is green. Most tomato plants have 'cut' leaves, but some have so-called 'potato' leaves (see Fig. 6.1).

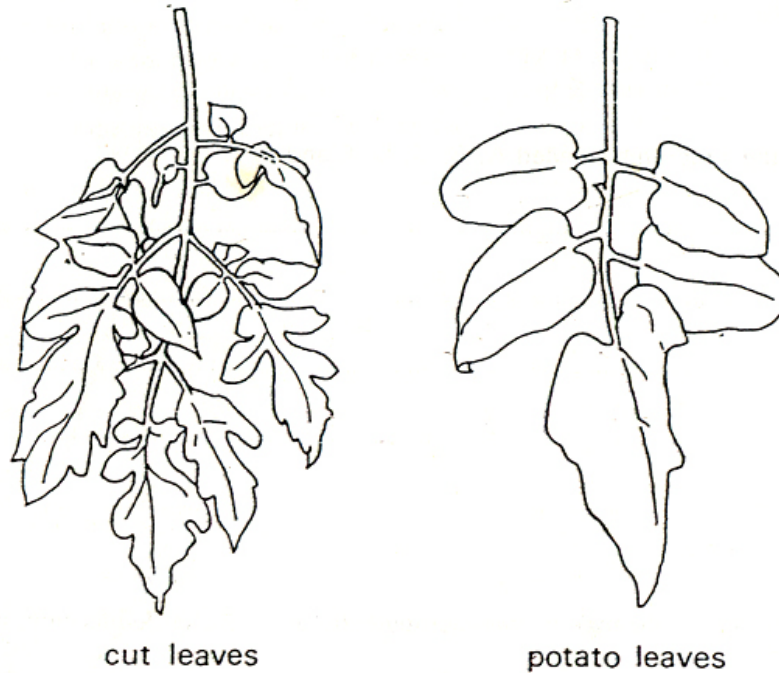


Fig. 6.1

Table 6.1 shows the numbers and phenotypes of offspring, resulting from two separate crosses of tomato plants:

<i>Parental phenotypes</i>		<i>Number and phenotypes of offspring</i>			
		<i>purple stem, cut leaves</i>	<i>purple stem, potato leaves</i>	<i>green stem, cut leaves</i>	<i>green stem, potato leaves</i>
(i)	purple stem, cut leaves X green stem, potato leaves.	354	0	367	0
(ii)	purple stem, cut leaves X green stem, cut leaves	693	241	0	0

Table 6.1

- (a) Deduce which of each pair of phenotypic characteristics is dominant, giving your reasoning.

Stem colour.

Dominant characteristic is *purple stem* [1]

Reasoning: [1]

- *When tomato plants with purple stem, cut leaves are crossed in (ii) with tomato plants with green stem, cut leaves, all the offspring have purple stem*
 - *This shows that the allele coding for purple stem expresses itself even in the presence of the allele for green stem*
-

Leaf shape.

Dominant characteristic is *cut leaves* [1]

Reasoning: [1]

- *When tomato plants with purple stem, cut leaves are crossed in (i) with tomato plants with green stem, potato leaves, all the offspring have cut leaves*
 - *This shows that the allele coding for cut leaves expresses itself even in the presence of the allele for potato leaves*
-

- (b) Deduce the probable genotypes for the parents in crosses (i) and (ii). Give your reasoning in each case, and use the symbols **A** / **a** for the dominant and recessive alleles respectively for stem colour, and the symbols **B** / **b** for the dominant and recessive alleles respectively for leaf shape.

Cross (i).

Parental genotypes: **AaBB, aabb** [1]

Reasoning: [2]

- **Both green stem and potato leaves are recessive characteristics**
- **The parental tomato plant with green stem, potato leaves must be homozygous recessive (genotype: aabb) for both characters**
- **F₁ phenotypic ratio = purple stem, cut leaves : green stem, cut leaves = 354 : 367 ≈ 1 : 1**
- **The parental tomato plant with purple stem, cut leaves must be homozygous dominant for one character but heterozygous for the other**
- **Since all the offspring have cut leaves (dominant characteristic for leaf shape), the genotype of the parental tomato plant with purple stem, cut leaves must be homozygous for leaf shape (i.e. BB)**
- **Since not all the offspring have purple stem (dominant characteristic for stem colour), the genotype of the parental tomato plant with purple stem, cut leaves must be heterozygous for stem colour (i.e. Aa)**

Cross (ii).

Parental genotypes: **AABb, aaBb** [1]

Reasoning: [2]

- **Green stem is a recessive characteristic**
- **The parental tomato plant with green stem, cut leaves must be homozygous recessive (genotype: aa) for stem colour**
- **F₁ phenotypic ratio = purple stem, cut leaves : purple stem, potato leaves = 693 : 241 ≈ 3 : 1**
- **Since all the offspring have purple stem (dominant characteristic for stem colour), the genotype of the parental tomato plant with purple stem, cut leaves must be homozygous for stem colour (i.e. AA)**
- **Since offspring with potato leaves (recessive characteristic for leaf shape) appeared even though both parental tomato plants have cut leaves (dominant characteristic for leaf shape), the genotype of both parental tomato plants must be heterozygous for leaf shape (i.e. Bb)**

(c) In some cases, the environment may affect the phenotype.

Give **one** named example to illustrate such an environmental effect.

[2]

- *Valid character – fur colour / synthesis of black pigment*
- *Matching type of organism – Himalayan rabbit*
- *Change in environment – exposure to heat*
- *Corresponding change in phenotype – white fur formed instead of black fur / no synthesis of black pigment*

[Total: 12]

- 7 Mammals represent a class of vertebrates known as Mammalia. There are about 5,500 known species of mammals, which are distributed into various taxa. Diprotodontia is a large order of about 120 marsupial mammals including kangaroos, wallabies, koalas, wombats, and many others. The genus *Macropus* in the family Macropodidae, is a group of marsupials commonly known as wallabies. Two of the 16 species of wallabies, with the specific epithet, *agilis* and *rufogriseus*, are most closely related to the kangaroos and confined to Hobart Island, near mainland Australia.

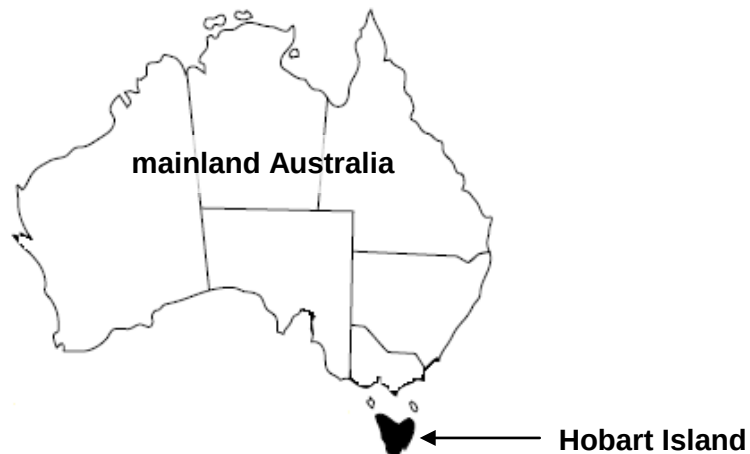


Fig. 7.1

- (a) Using the information above, give an example of each of the following:

- (i) the kingdom in which wallabies are classified

• *Animalia / Animal*

- (ii) the name of an order

• *Diprotodontia*

- (iii) the name of a family

• *Macropodidae*

- (iv) a binomial name

• *Macropus agilis* OR *Macropus rufogriseus*

[2]

- (b) Suggest why the populations of wallabies stated in (a)(iv) are classified as a single species. [1]

- *Similar morphological / anatomical / physiological features / ecological niche*
- *Capable of inter-breeding*
- *Shares a common ancestor*
- *Produce viable and fertile offspring*

- (c) (i) Describe how the evolutionary relationship between wallabies and kangaroos can be best determined. [2]

- *(Example) Cytochrome c*
- *Evaluating the molecular homologies / molecular analysis based on DNA / amino acid comparison*
- *Sequence alignment of comparable sequence is to be carried out*
- *Two species considered to be closely related should have a greater proportion of their DNA in common*
- *Inherited from a common ancestor*

- (ii) State **one** advantage of the method suggested in (c)(i). [1]

- *Enable the understanding of phylogenetic relationships that cannot be determined by non-molecular methods (i.e. comparative anatomy)*

- (d) Explain how the 16 closely related species of wallabies may have arisen. [3]

- *Different environment act as different selection pressures*
- *Natural selection*
- *Wallabies at a selective advantage will have a higher probability of surviving and reproducing*
- *Advantageous genes (alleles) will be passed on to the next generation*
- *Alters allele frequency*
- *Proportions of populations exhibiting favorable traits will increase*
- *Geographical isolation*
- *Reproductive isolation*
- *Gene flow between populations of existing species is disrupted*
- *Interbreeding is prevented*
- *Separated gene pool may diverge through mechanisms such as mutations and genetic drift*

In June 2006, a heavy rainfall had caused a severe outbreak of melioidosis (a bacterial disease that affects many species of animals including wallabies), which led to a high death toll among the wallabies that are confined to Hobart Island.

(e) (i) Name the evolutionary event that has occurred. [1]

- *Bottleneck effect*

(ii) Discuss the effects of (e)(i) on the populations of wallabies in Hobart Island. [2]

- *A drastic reduction in the size of wallabies population*
- *Changes in the allele frequency*
- *Decrease in genetic diversity*
- *Continuous inbreeding will occur*
- *Decrease in relative fitness*

[Total: 12]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
 Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
 Your answer must be in continuous prose, where appropriate.
 Your answer must be set out in sections (a), (b), etc., as indicated in the question.

- 8 (a)** Define control elements and explain how they influence transcription in the eukaryotic genome. [7]

(Definition)

- *Control elements are segments of non-coding DNA that help to regulate transcription by binding to transcription factors or to RNA polymerase*

(Promoters)

- *Promoters bind to RNA polymerase*
- *RNA polymerase / general transcription factors (GTFs) recognizes the TATA box within the promoter*
- *Promoter may also bind to general transcription factors (GTFs) to form a transcription initiation complex*
- *When RNA polymerase and TFs bound, allows for transcription*
- *General transcription factors (GTFs) bound to promoter or proximal control elements mediate the binding of RNA polymerase to the promoter*
- *Interaction of promoter or proximal control elements to GTFs and GTFs interaction with RNA polymerase usually leads to only a basal level of transcription / production of a little RNA*

(Enhancers & Silencers – in general)

- *Enhancers and silencers bind to specific transcription factors*

EITHER

- *Enhancers bind to activators to upregulate transcription / increase transcription*
- *Silencers bind to repressors to inhibit or downregulate transcription / stop transcription*

OR

- *Enhancers or silencers bound to specific transcription factors (STFs) allow for the regulation of the level of gene expression*

(Distal control elements – Enhancers)

- *Distal control elements / some enhancers and silencers are located a long distance away from the promoter / thousands of nucleotides upstream or downstream of the promoter*
- *When enhancer (distal) binds with an activator, it causes the binding of a DNA-bending protein*
- *This brings the bound activator / STF closer to the promoter / Enables the enhancer to influence a promoter hundreds or even thousands of nucleotides away*
- *Activators then bind to certain GTFs and mediator proteins / these multiple protein-protein interactions help assemble and position the transcription initiation complex on the promoter / allow RNA polymerase to bind more efficiently or with higher affinity*

- *Enhancers binding to activators may cause histone acetylation / increase gene accessibility to RNA polymerase and TFs / decondense DNA*
- *Increasing the rate of transcription*

(Distal control elements – Silencers)

- *Silencers bound to repressors can also inhibit RNA polymerase binding*
- *Silencers bound to repressors may cause histone deacetylation / decrease gene accessibility to RNA polymerase and TFs / condensing DNA*
- *Decreasing the rate of transcription*

(Combinatorial control)

- *Combinatorial control of genes occurs because each gene has a unique combination of CE*
- *Depending on the different types of TF / activators found in each cell, genes will be switched on*
- *Determines cell-type specific transcription*

(Coordinate control)

- *Coordinate control of genes with related function occurs because the group of genes share the unique combination of CE*

- (b)** Describe how differences in the structure and organization of prokaryotic and eukaryotic genomes affect their control of gene expression. [7]

		<i>Prokaryotic Genome</i>	<i>Eukaryotic Genome</i>
<i>Structure</i>	<i>Level of condensation</i>	• <i>DNA not highly condensed / Not bound to histones</i> • <i>Because genome size is smaller</i>	• <i>Many levels of condensation of DNA / DNA bound to histones</i> • <i>Because genome size is larger</i>
	<i>Significance</i>	• <i>No need to decondense chromosome / uncoil DNA from histones before transcription can occur</i>	• <i>Chromatin must be decondensed / have a looser conformation for transcription to occur so that RNA polymerase & transcription factors can bind to DNA</i>
		• <i>Changes in chromosome structure not used as method to regulate transcription</i>	• <i>Rate of transcription can be regulated by histone acetylation</i> • <i>resulting in conversion between euchromatin and heterochromatin, affecting the ease of transcription</i>
<i>Structure / Organisation</i>	<i>Chromosomes / Genomes bound by membrane</i>	• <i>Not bound by membrane / genome in nucleoid</i>	• <i>Bound by the nuclear membrane / genome in nucleus</i>
	<i>Significance</i>	• <i>Transcription and translation occur simultaneously</i>	• <i>Transcription occurs before translation / transcription occurs within the nucleus and</i>

			<i>translation outside the nucleus / transcription and translation occur in different locations in the cell</i>
		<i>Post-transcriptional control of gene expression not possible</i> <i>Thus mRNA is more easily degraded / less stable</i>	<i>Post transcriptional control of gene expression is possible</i> <i>Eg. Alternative splicing / mRNA modifications before translation – 5' cap or poly-A tail</i> <i>Thus mRNA is less easily degraded / more stable</i>
Structure / Organisation	Presence / Absence of introns	<i>No introns</i>	<i>Presence of introns</i>
	Need for mRNA splicing & allows for alternative splicing	<i>No splicing of mRNA is needed</i> <i>No alternative splicing possible</i> <i>Only one mRNA / protein product per gene</i>	<i>Splicing of pre-mRNA is needed / exons / coding regions must be spliced together</i> <i>Allows for alternative splicing</i> <i>giving rise to varying mature mRNA molecules and hence multiple possible protein variants from the same gene</i>
Organisation	Presence / absence of operon	<i>Genes coding for enzymes with related function grouped in an operon</i>	<i>Dispersed genes coding for enzymes with related function each controlled by an individual / separate promoter</i>
	Need for coordinate control, eukary genes having shared control elements	<i>Produces polycistronic mRNA</i> <i>Allows coordinated control of metabolic enzymes involved in the same pathway</i>	<i>Produces monocistronic mRNA</i> <i>Coordinated control occurs via binding of transcription factors to control elements shared by genes with related functions (when genes are on diff chrom)</i> <i>Coordinated control also occurs by DNA methylation / histone acetylation that causes all the genes in the same region to be made</i>

			<i>available or unavailable for transcription (when genes w related fns are close together on the same chrom)</i>
		<i>All gene products of an operon are produced in the same amount</i>	<i>Each gene product can be produced in different amounts or not at all</i>
Organisation	Presence / Absence of (distal) control elements	<i>No distal control elements or its regulatory sequences are close to promoter</i>	<i>Presence of distal control elements</i>
	<i>Allows for more fine-tuned control of transcription</i>	<i>Simple control possible, either transcription switched on or off</i> <i>Level of expression cannot be finely tuned / regulated as specifically</i>	<i>Allows for fine tuned control of gene expression / allows for different levels of expression of a particular gene depending on the activators and repressors present in the cell at a specific time</i>

(c) Describe the eukaryotic processing of pre-mRNA.

[6]

{5' cap}

{Description}

- Addition of 5' cap*
- After transcription / 1st of the 3 processes to take place*
- Catalysed by a capping enzyme*
- 7-methylguanosine added / modified guanine*

{Function}

- Facilitates the export of mRNA from nucleus to cytoplasm / through nuclear membrane*
- Small subunit of ribosome may bind to it to initiate translation*
- Protects RNA transcript from degradation by exonucleases*

{poly(A) tails}

{Description}

- *Endonuclease recognizes polyadenylation sequence*
- *Endonuclease cuts pre-mRNA several nucleotides beyond 3' end of polyadenylation sequence*
- *Addition of poly(A) tails to 3' ends / polyadenylation to 3' ends*
- *Catalysed by poly-A polymerase*
- *Poly(A) tails can be about 200 nucleotides long*

{Function}

- *Protects mRNA from degradation by exonuclease*
- *Facilitates the export of mRNA from nucleus to cytoplasm / through nuclear membrane*

{mRNA splicing}

{Description}

- *mRNA splicing*
- *introns / intervening sequences are spliced out by the action of a spliceosome*
- *pre-mRNA can be spliced in different ways / alternative mRNA splicing*
- *Introns can be removed separately or in combination*
- *If two successive introns are removed together, the exon between them will be removed as well / opportunity to modify the coding sequence of RNA by deleting some of its exons*
- *Alternative mRNA splicing allows a single gene to encode different polypeptides / different RNA transcripts to be produced by a single gene / protein variants from a single gene*

- 9 (a) Outline the importance of mitosis and explain the need to keep cell division under fine control. [7]

(IMPORTANCE OF MITOSIS)

- *Mitosis results in genetically identical daughter cells*
- *To maintain genetic stability of an organism*
- *Each daughter cell has the same number of chromosomes and hereditary information as the parental cell*
- *To enable growth and development of a multicellular organism*
- *Net increase in the number of cells*
- *To enable tissue repair / renewal*
- *Worn-out or damaged cells replaced by new cells that are identical to the original cells*
- *Basis of asexual reproduction*
- *Only one parent / organism is involved in producing new individual organisms*
- *e.g. binary fission in Amoeba and development of a new potato plant from a tuber of a potato plant*

(THE NEED TO KEEP CELL DIVISION UNDER FINE CONTROL)

(General function of cell cycle checkpoints)

- *Cell division is finely controlled by multiple cell cycle checkpoints*
- *that verify whether the processes at each phase of the cell cycle have been accurately completed before progression into the next phase*
- *Defective cell cycle checkpoints can lead to tumour formation / cancer*

(G₁ / restriction checkpoint)

- *Should the cell enter S phase in the absence of mitogens / growth factors and before the minimum cell size is reached (i.e. defective G₁ / restriction checkpoint)*
- *cell division is said to be deregulated / independent of environmental conditions and subsequent phases of the cell cycle (e.g. DNA replication, mitosis) might not occur properly due to limited cell size*

(G₂ checkpoint)

- *Should the cell enter M phase even when not all genes have been replicated (i.e. defective G₂ checkpoint)*
- *some genes would be missing in the daughter cells*

(M checkpoint)

- *Should the cell transit into anaphase even when not all chromosomes are aligned properly at metaphase (i.e. defective M checkpoint)*
- *aneuploidy or polyploidy would result*

- (b) Describe the development of cancer as a multi-step process and suggest ways in which retroviral infection may increase the chance of cancer development. [7]

(DEVELOPMENT OF CANCER AS A MULTI-STEP PROCESS)

- *More than one somatic mutation is generally needed to produce all the changes characteristic of a full-fledged cancer cell*
- *The longer we live / The older we get, the more likely we are to accumulate cancer-causing mutations*
- *that result in the gain of function mutation in at least 1 proto-oncogene*
- *and the loss of function mutation in several tumour suppressor genes*
- *Furthermore, since mutant tumour suppressor alleles are usually recessive, mutations must knock out both alleles in a cell's genome before the function of the tumour suppressor is lost*
- *Most oncogenes, on the other hand, behave as dominant alleles*
- *Finally, in many malignant tumours, the gene for telomerase is activated*
- *Production of telomerase in cancer cells removes a natural limit on the number of times the cells can divide*
- *More mutations are required for the tumour cells to metastasize*

(WAYS IN WHICH RETROVIRAL INFECTION MAY INCREASE THE CHANCE OF CANCER DEVELOPMENT)

- *When the viral genome inserts into a particular site in the host chromosome, it may inactivate / disrupt a tumour suppressor gene of the host cell*
- *Viral infection may thus increase the chance of getting cancer by causing loss of function mutations in tumour suppressor genes / tumour suppressor proteins to lose their normal functions*

OR

- *A cellular proto-oncogene may also be activated / over expressed*
- *when the viral genome inserts near the cellular proto-oncogene, which is now under the control of the viral promoter / enhancer*
- *Viral infection may thus increase the chance of getting cancer by causing gain of function mutations in proto-oncogenes / proteins that stimulate cell division to gain in function / oncogenes to form*

- (c) Propose **three** possible strategies that doctors may use to fight influenza viral infections at the molecular level. [6]

- *Inhibit the attachment and thereby entry of the virus into its host cell by developing antibodies that bind to the virus or its receptors on the host cell's plasma membrane*

OR

- *Inhibit the synthesis of viral genomes inside the host cell by developing nucleotide or nucleoside analogues that look like the building blocks of RNA but deactivate the enzymes that synthesize the nucleic acid once the analogue is incorporated*

OR

- *Inhibit the operation of the viral genomes by developing anti-sense RNA or DNA whose sequence is complementary to that of critical sections of the viral genomes*

----- The End -----